

Technical Appendix

Reproducibility Scope and Data

The evaluation uses the complete named public populations described in the main paper. No novel benchmark dataset is introduced. The local coding-agent histories are observational case-study inputs rather than a new dataset, and conclusions drawn from them are limited to the reported populations. Public benchmark targets, human stages, outcomes, and scores are used only by scorers after each method’s output is fixed.

The paper makes no theoretical contribution; the theoretical-contribution items in the reproducibility checklist therefore do not apply. AGENTPROF is a 9.8K-LOC implementation, so executable source, preprocessing scripts, and released-data manifests belong in the separately submitted code-and-data package rather than as printed source listings in this appendix.

Automatic Segmentation and Identity

The direct annotation backend is the OpenAI Codex CLI (version 0.145.0, model `gpt-5.6-sol`, high reasoning effort, default decoding, sandboxed non-interactive mode). Each worker receives one source-only trajectory packet, makes one annotation call with at most one format retry, and shares no state with other workers. Prompt and response previews are truncated to 1,800 characters, tool commands to 600, and status previews to 200; additive measures come from structured counters and are not truncated. The fixed instruction is reproduced in the main paper. It emits sparse path changes whose names begin with a verb and contain one to three words; intervening steps inherit the preceding path.

Before folding, one fixed source-only action-object map canonicalizes free-form interval names. If canonicalization would make adjacent display paths equal, a boundary-safe refinement retains the minimum source words needed to distinguish them; unresolved collisions fail closed. Equal canonical names receive one stable operation ID across sessions. On CodeTraceBench, this maps 3,895 free-form names to 783 stable names with zero adjacent display-path collisions while preserving every temporal mark.

The non-LLM recurrence backend counts adjacent action transitions in reference sessions and scores each transition by normalized pointwise mutual information (NPMI),

$$\text{NPMI}(a, b) = \frac{\ln[p(a, b)/(p_L(a)p_R(b))]}{-\ln p(a, b)},$$

where all probabilities use the same transition sample space (Bouma 2009). Occurrence-weighted one-dimensional k -means with $k = 2$ initializes at the minimum and maximum scores, assigns distance ties to the lower center, and uses the converged midpoint as the cutoff (MacQueen 1967). The same deterministic procedure is applied to action-changing transitions. Same-action pairs use the global cutoff; different-action pairs use the smaller cutoff. A detailed (`action`, `action_detail`) arm may remove a coarse boundary but never add one; missing or unseen detail falls back to the coarse decision.

RQ2 Protocol

Within each workload, operations, target-blind paths, benchmark judge/localizer predictions, and scoring rules are fixed before test targets are loaded. AgentProcessBench averages frozen judge votes within a group. HINTBench and TraceElephant assign each root-to-frame prefix the 95% Wilson lower bound of its frozen positive-prediction fraction; an operation receives the maximum score over prefixes containing it. The released HINTBench snapshot contains 536 of the paper-reported 629 trajectories. A separate 80-trajectory validation snapshot selects one of 24 field orders before the 536 test trajectories are scored.

Each target-bearing trajectory contributes one AP value, and MAP is their arithmetic mean. The reported `Direct+AGENTPROF-minus-Direct-only` and `Direct+AGENTPROF-minus-Direct+AGENTPROF-Raw` intervals use 10,000 paired resamples of trajectory clusters within benchmark-defined strata. Zero-positive trajectories are retained for population coverage but excluded because AP is undefined without a relevant item.

For the TraceElephant reading study, the Grok 4.5 CLI reader is invoked with tools and subagents disabled. Full-trace reading uses one single-turn call per query. Profile-guided reading first exposes only the target-blind operation skeleton; the reader selects at most five groups, after which a second single-turn call receives only those groups’ source-visible evidence. Unranked operations are appended in original order. Semantic and raw-action conditions differ only in the operation names used by the skeleton.

RQ3 Protocol

For CodeTraceBench, all 405 annotations are materialized before the 2,948 human stage labels are loaded by the scorer. Codex receives prompts, commands, and output summaries, but no label or score. It emits 4,496 sparse marks over 17,148 source-native steps. B^3 scores the resulting operation partition, and exact adjacent-boundary precision, recall, and F1 score the transition decisions. Paired task-cluster intervals preserve the benchmark’s task grouping.

The OSWorld-Human study uses all 287 sessions, containing 3,978 operations, 3,691 adjacent pairs, and 2,042 human groups. The supervised Bernoulli Naive Bayes predictor (McCallum and Nigam 1998) fixes its model family, nine input fields, adjacent-pair features, and training procedure. Five session-held-out folds fit parameters and the decision threshold on training sessions and predict each held-out session exactly once. Label-free recurrence uses only visible transitions from the other four folds. Reference-calibrated recurrence additionally fits one scalar NPMI cutoff using the training-fold group annotations; it never reads held-out groups.

The evaluation-only Qwen3.6-27B llama.cpp backend classifies all 1,012 AgentBoard goals from goal text and the nine declared family descriptions. Three complete runs produce identical assignments. The action-label adapter uses the fixed eight TraceView action definitions and the current thought/action only; two complete runs over all 2,737 labels produce identical assignments. Literal labels use accuracy

and macro-F1, partitions use V-measure or ordinary B^3 , and transitions use exact adjacent-boundary precision, recall, and F1.

Runs, Uncertainty, and Computing Environment

The CodeTraceBench direct backend makes one annotation call per trajectory apart from the stated format retry. RQ2 fixes one profile and one benchmark prediction per trajectory, then computes uncertainty with 10,000 paired resamples. The TraceElephant reader makes one call per stage and query. OSWorld-Human uses five held-out folds. The closed-label repeat counts are given above. RQ4 construction times are medians of three complete runs; peak RSS is the largest observed semantic-profile peak. Once marks are fixed, projection, folding, serialization, and pprof readback are deterministic.

The RQ4 construction measurements use `agentpprof 0.2.37` on a 24-core Intel Core Ultra 9 285K with 125 GiB RAM and Linux 6.15.11. The 27,765-operation union completes semantic construction in 1.16 s at 465.2 MiB peak RSS, versus 0.97 s for raw-action grouping. The automatic CodeTraceBench pass consumes 12,050,384 input and 231,886 output tokens and 2,215.858 s of active backend time. Deterministic materialization of the 440-trajectory population takes 0.26 s for operation weights and 0.25 s for token weights.

Privacy, Ethics, and Responsible Use

Agent histories can contain prompts, model responses, source code, file paths, commands, network targets, and system-effect records. Semantic operation names and pprof labels are derived data, but they can still reveal project intent, behavior, or identity. The local histories in the case studies are observational inputs, not a released benchmark; the paper reports aggregate measurements and task-level paths rather than publishing their raw histories. A reproduction package should exclude private histories and use the cited public populations for repeatable experiments.

Automatic annotation adds a separate disclosure boundary because a model backend reads source packets. Preview length limits bound packet size but do not redact secrets. Before using a remote backend, operators should remove credentials and personal data, minimize fields, and select a provider and retention policy appropriate for the data; sensitive workloads can instead use an approved local backend or the non-LLM recurrence backend. Profiles, annotations, and intermediate packets should receive the same access control and retention protection as their source traces.

Semantic boundaries and names reflect the selected backend and instruction, not objective or causal judgments about a person or agent. Agreement with human stages measures structural correspondence on the named datasets, not universal correctness. Profiles should therefore guide inspection rather than autonomously determine employment, access-control, or safety decisions. Likewise, process, file, and network capture should be limited to systems and trajectories the operator is authorized to observe. Using cross-run profiling for covert worker surveillance or to infer unrelated private

activity falls outside the intended debugging and resource-analysis purpose.

Finally, public benchmarks and model backends can carry task, language, and annotation biases. Results are bounded to the reported populations and should be revalidated before deployment in a materially different setting. Model annotation also incurs the measured token and compute cost in RQ4; reusing fixed annotations and choosing the non-LLM backend where its accuracy is adequate reduce repeated inference.

Reproducibility Checklist Notes

Checklist items concerning novel datasets are not applicable because the paper introduces no benchmark dataset. Items concerning theoretical claims are likewise not applicable. This appendix supplies the final algorithm settings, target-blind freezing rules, evaluation metrics, run counts, and the measured RQ4 environment. Full preprocessing and experiment source should be assessed from the separately submitted code-and-data package; a PDF appendix cannot by itself satisfy source-code availability items.